

Project Milestone 3

Milestone	Brief Description	Deadline
3: SA (6%)	Implementation of Simulated Annealing (SA) algorithm on your specific problem.	Week 6: Thursday, 11 November 2021

Milestone 3 Objectives:

1. Simulated Annealing (SA) Implementation:

- Based on the problem formulation, implement SA for your project topic.
- Visualize the resulting solutions via illustrative figures.
- Validate the algorithm by testing on case studies and comparing to results from previous literature or known benchmark problems tackling the same problem.
- Test the algorithm on different case studies of varying sizes and complexity.

2. Progress Report:

- Explain and illustrate your implementation of the algorithm for your problem (e.g. via algorithm pseudocode or flowchart); including problem-specific considerations (e.g. neighborhood operators).
- Describe all case studies and specify the used input parameters for each case study.
- Report and discuss the results of your algorithm validation.
- Report and discuss all obtained results of each case study.
- Analyze and discuss the algorithm's performance and the effect of the different algorithm parameters on its performance.

Programming Languages:

- The programming language can be either **MATLAB or Python**.
- The chosen language **will not be changed** after this milestone submission.
- In case Python was the chosen programming language, you may use MATLAB only for generation of graphs, figures, or charts.
- Your codes should be runnable on any machine (i.e. usage of toolboxes/packages is not allowed).

Minimum Requirements:

- Implementation of SA on your problem.
- Visualization of resulting solutions.
- Validation of results.
- Discussion and analysis of results via different methods.

Deliverables:

1. Progress report (in provided template) in accordance to objectives above.
2. Your implemented codes for SA and for all case studies and results.
3. Screen-recorded video of your codes running and the codes' results including the visualization of the results with narrated comments.

Submission Deadline:

- The deadline of Milestone 3 submission is on **Thursday, 11 November 2021, at 23:59**.

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The milestone will be graded based on several criteria including but not limited to:

- Quality of all submitted material.
- For codes implementation: scalability, modularity, documentation (i.e. well commented), etc.
- For results analysis and discussion: analysis of results (numerical, graphical, performance metrics, parametric studies, statistical studies, etc.) in light of the specific problem on-hand; as well as commenting on noticed trends and ability to highlight recommendations for improvement.
- For reporting: clarity of presentation and professional discussion with multiple aids (e.g. block diagrams, flowcharts, figures, tables, graphs, charts, etc.). Equations should be written professionally with proper identification of nomenclature.